

Article

A Distributed Energy Storage-Based Planning Method for Enhancing Distribution Network Resilience

Yitong Chen ¹, Qinlin Shi ¹, Bo Tang ^{1,*}, Yu Zhang ² and Haojing Wang ²

¹ Faculty of Electrical Engineering, Shanghai University of Electric Power, Yangpu District, Shanghai 200090, China; 18017347618@163.com (Y.C.); sshiep0920@126.com (Q.S.)

² State Grid Shanghai Electric Power Company Electric Power Science Research Institute, Hongkou District, Shanghai 200437, China; yxsgcc20@mail.shiep.edu.cn (Y.Z.); sunhang@mail.shiep.edu.cn (H.W.)

* Correspondence: tangbo@shiep.edu.cn

Abstract

With the widespread adoption of renewable energy, distribution grids face increasing challenges in efficiency, safety, and economic performance due to stochastic generation and fluctuating load demand. Traditional operational models often exhibit limited adaptability, weak coordination, and insufficient holistic optimization, particularly in early-/mid-stage distribution planning where feeder-level network information may be incomplete. Accordingly, this study adopts a planning-oriented formulation and proposes a distributed energy storage system (DESS) planning strategy to enhance distribution network resilience under high uncertainty. First, representative wind and photovoltaic (PV) scenarios are generated using an improved Gaussian Mixture Model (GMM) to characterize source-side uncertainty. Based on a grid-based network partition, a priority index model is developed to quantify regional storage demand using quality- and efficiency-oriented indicators, enabling the screening and ranking of candidate DESS locations. A mixed-integer linear multi-objective optimization model is then formulated to coordinate lifecycle economics, operational benefits, and technical constraints, and a sequential connection strategy is employed to align storage deployment with load-balancing requirements. Furthermore, a node-block-grid multi-dimensional evaluation framework is introduced to assess resilience enhancement from node-, block-, and grid-level perspectives. A case study on a Zhejiang Province distribution grid—selected for its diversified load characteristics and the availability of detailed historical wind/PV and load-category data—validates the proposed method. The planning and optimization process is implemented in Python and solved using the Gurobi optimizer. Results demonstrate that, with only a 4% increase in investment cost, the proposed strategy improves critical-node stability by 27%, enhances block-level matching by 88%, increases quality-demand satisfaction by 68%, and improves grid-wide coordination uniformity by 324%. The proposed framework provides a practical and systematic approach to strengthening resilient operation in distribution networks.



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Keywords: distributed energy storage systems; distribution network operation optimization; priority indices; sequential configuration; multi-dimensional evaluation framework

1. Introduction

1.1. Motivation

High penetration of distributed generation (DG) and flexible loads is transforming distribution networks from passive, unidirectional systems into active, bidirectional “generation-network-load-storage” systems [1]. However, the resulting uncertainties and fluctuations

on both generation and load sides challenge grid stability and efficiency. Traditional methods—such as grid reinforcement, demand-side management, and conventional reactive compensation—lack the speed and adaptability to address real-time imbalances from renewable intermittency. Insufficient resilience can lead to voltage and frequency deviations, cascading failures, and even large-scale blackouts with severe economic impacts [2,3].

In this context, Distributed Energy Storage Systems (DESS) have become crucial. With their dual “generation-load” capabilities and fast response times, DESS offer vital services like dynamic power balancing, rapid frequency and voltage regulation, and post-fault recovery—functions that traditional solutions cannot fully provide [4,5].

At the distribution level, the increasing penetration of renewable generation introduces strong uncertainty in both local operating conditions and load evolution, particularly in the early and intermediate stages of energy transition. Under such circumstances, planning decisions for distributed energy storage are increasingly oriented toward enhancing resilience, power quality, and operational adaptability within a given planning horizon and investment scale [6]. This perspective motivates the need for demand-aware and flexible planning approaches for distributed energy storage deployment.

The rapid growth of China’s energy storage sector highlights the urgency of this transition. In 2023, China added 21.5 GW of new energy storage capacity, nearly half of the global total. By 2030, cumulative installed capacity is projected to exceed 200 GW [7]. This growth is driven by varying regional needs: urban grids, where supply reliability is essential despite limited DG penetration, and rural grids, where excess renewable generation often outpaces local consumption [8]. Deploying DESS is therefore a critical strategy for enhancing distribution network resilience, facilitating renewable integration, and ensuring secure, efficient grid operation amidst high DG penetration.

1.2. Literature Review and Research Gaps

Distribution networks are facing challenges such as power imbalance and voltage violations caused by renewable energy fluctuations, making the deployment of DESS for dynamic regulation an urgent need, as evidenced by existing research. Reference [9] introduces a two-tier optimization framework utilizing cluster partitioning to develop an upper-level capacity planning model and a lower-level node scheduling model. Reference [10] describes a configuration approach that uses spectral clustering partitioning and two-tier optimization to promote power complementarity within clusters and improve overall system stability by optimizing the outer layer’s location and capacity determination as well as the inner layer’s output adjustments. References [11,12] propose DESS-based coordinated voltage regulation strategies for distribution networks with high renewable penetration, where DESS are explicitly modeled and coordinated through hierarchical or multi-level control frameworks to mitigate voltage violations. Reference [13] focuses on the optimal allocation of distributed modular energy storage systems for voltage regulation, aiming to minimize voltage deviations by determining the optimal locations and capacities of DESS units in active low-voltage distribution networks. Nonetheless, while many existing centralized or cluster-based DESS architectures are effective in maintaining regional power equilibrium and voltage regulation, their reliance on simplified or deterministic representations of load dynamics and renewable generation uncertainty may undermine operational resilience, leading to storage over-allocation, reduced utilization efficiency, and suboptimal economic performance in practical applications.

Reference [14] establishes a bi-level coordinated siting and sizing planning model for DG and DESS, selecting interconnection points with the objective of minimizing line and operational investment costs, where system-level planning decisions are made without explicitly evaluating node-level priority or sensitivity characteristics. In references [15,16],

node voltage sensitivity is utilized to assess the voltage regulation needs of various nodes, and the optimal placement of DESS is determined through sequential optimization. Subsequent studies have extended this voltage-sensitivity-based paradigm by constructing more comprehensive node sensitivity indices to improve the effectiveness of sequential DESS siting and sizing under typical operating conditions [17]. References [18,19] focus on optimizing the charging and discharging power as well as the capacity of DESS for unbalanced distribution networks with a significant penetration of DG, using node loss sensitivity variance as a key metric. More recent work has further refined loss-based sensitivity analysis by defining integrated power-flow sensitivity variance indices and adopting staged siting–sizing procedures, while still relying on deterministic or representative operating scenarios [20]. Nevertheless, most existing DESS planning approaches based on node prioritization and sequential deployment are developed under well-defined network topology and load characteristics, often considering single or representative scenarios and limited functional objectives. Such assumptions may lead to complex, redundant, and economically inefficient configurations when applied to large-scale distribution networks with heterogeneous source–load characteristics. Meanwhile, existing research largely focuses on strategic expansion planning with detailed power-flow modeling, whereas tactical, planning-oriented DESS deployment under incomplete network information—where prioritization, deployment order, and multi-scale resilience assessment are critical—has received relatively limited attention. To address this gap, this paper proposes a planning-oriented DESS deployment framework that integrates demand-driven prioritization, priority-guided sequential planning, and node–block–grid multi-scale resilience evaluation, aiming to enhance day-to-day operational resilience of distribution networks under uncertainty. It is emphasized that the proposed framework does not aim to replace detailed capacity expansion or operational models; rather, it provides a complementary decision layer that can inform DESS siting, sizing, and deployment order, and can be readily incorporated into more comprehensive planning or operational formulations when detailed network models are available.

With the increasing penetration of photovoltaic and wind power in distribution networks, the modeling of renewable generation uncertainty has become an important issue for planning-oriented studies. Existing review works generally classify scenario generation methods into explicit statistical approaches and implicit deep generative approaches. Statistical methods are widely adopted in planning applications due to their interpretability and compatibility with optimization models, whereas deep generative methods are capable of capturing complex correlations but often require large datasets and lack transparency [21]. In this paper, a Gaussian Mixture Model (GMM) is employed to generate representative renewable power output scenarios. GMM is selected because it can effectively describe multimodal distributions and stochastic fluctuations of renewable generation, while maintaining clear physical interpretability through explicit model parameters. Moreover, existing studies have shown that GMM-based clustering and modeling approaches can achieve satisfactory performance in photovoltaic power characterization and typical scenario identification, supporting its suitability for planning-oriented applications [22]. At the same time, recent research has emphasized that the quality of generated scenarios should not be evaluated solely based on basic statistical moments, but also in terms of randomness and diversity [23]. Accordingly, the adopted GMM-based framework aims to provide representative and diverse scenarios that are well suited for long-term and medium-term DESS planning. The brief description of the literature are shown in Table 1.

1.3. Contributions and Paper Structure

This research provides a distributed energy storage optimization strategy to enhance the operational performance of distribution grids by innovatively integrating priority in-

trices with sequential allocation strategies. By conducting a thorough analysis of quality and efficiency demands, along with a multi-faceted evaluation, it directs the best distribution of energy storage to enhance the operational economy of distribution grids, facilitate the integration of renewable energy sources, and improve system balancing performance. The specific contributions are summarized as follows:

(1) A priority index model is proposed to quantify the demand for energy storage deployment at different nodes within a grid-based distribution network. Driven by multi-dimensional user demand characteristics, the model enables quantitative assessment of storage configuration priorities across nodes and provides a demand-oriented basis for subsequent DESS planning.

(2) Based on the grid-based hierarchical structure of distribution networks, a priority-guided sequential planning strategy for distributed energy storage systems is proposed under renewable uncertainty. Wind and photovoltaic output variability is characterized using a GMM. Guided by priority indices, the framework sequentially determines DESS siting and sizing to enhance distribution network resilience under multiple economic and operational objectives, thereby overcoming the limitations of one-shot planning methods.

(3) A multi-level evaluation framework is established to assess the impacts of distributed energy storage deployment on distribution network resilience from node-, block-, and grid-level perspectives. By jointly considering localized operational performance, regional coordination effects, and overall system behavior, the framework provides a comprehensive evaluation of DESS planning outcomes.

Table 1. A brief description of the literature.

Ref.	Network Scale/Structure	Node Priority Evaluation	Planning Strategy	Considered Objectives	Renewable Uncertainty Modeling	System-Level Evaluation
[8,9]	Cluster-based DN	No	Bi-level/two-tier planning	Cost, power complementarity	No	No
[10,11]	DN with high RES	No	DESS-based coordinated voltage regulation	Voltage regulation	No	No
[12]	System-level DN with DG–DESS	No	Bi-level coordinated expansion planning	Investment and operation cost	No	Yes
[13–15]	Unbalanced DN	Yes (voltage sensitivity)	Sequential placement	Voltage/loss	No	No
[16–18]	DN with high DG	Yes (loss sensitivity variance)	Sequential optimization	Loss, capacity	No	No
[19–22]	Wind/PV uncertainty modeling	No	Scenario generation and evaluation	Uncertainty characterization	Yes	No
This paper	Grid–block–node DN	Yes (multi-dimensional priority index)	Priority-guided sequential allocation	Economy, resilience, RES integration	Yes	Yes

To clearly illustrate the overall research logic and the interaction among the proposed priority index model, multi-objective optimization, and the node–block–grid resilience evaluation framework, the overall methodological framework of this study is presented in Figure 1. As shown in the figure, uncertainty modeling and data-driven indicators are first used to quantify regional energy storage demand, followed by coordinated optimization of DESS configuration, and finally evaluated through a multi-dimensional resilience assessment across node, block, and grid levels.

The remaining sections of this work are organised as follows. Section 2 describes the evaluation approach based on the vast distribution network prerequisites. Section 3

creates a priority index to guide differential energy storage distribution by assessing resilience requirements across multiple dimensions, such as power supply dependability and renewable energy integration. Section 4 describes the methodical planning technique for DESS implementation. Section 5 introduces a systematic evaluation framework for statistically assessing the effectiveness of distributed storage configurations in improving grid resilience. Section 6 concludes the study.

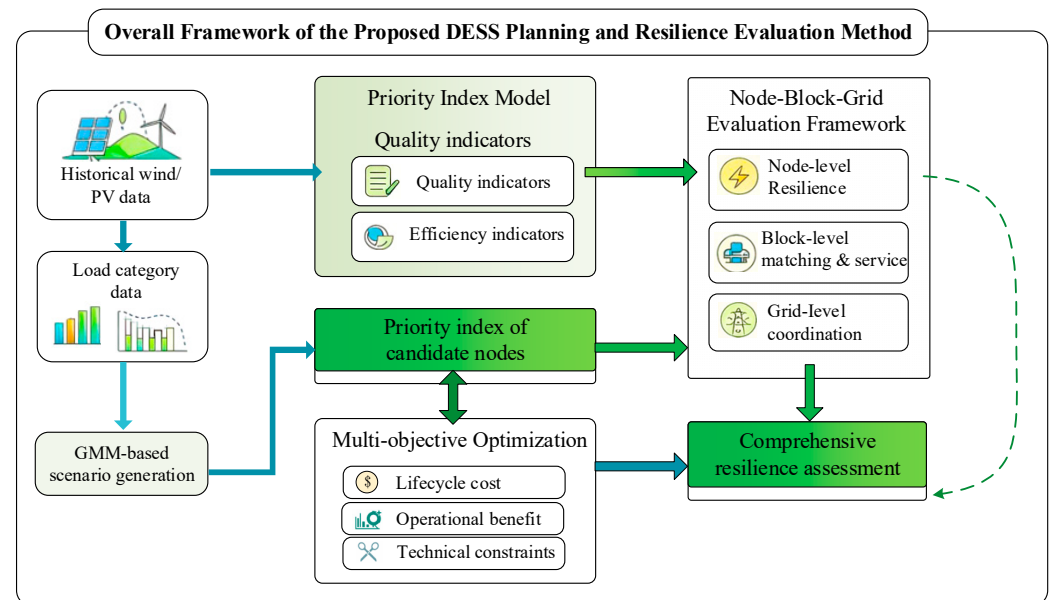


Figure 1. Overall framework of the proposed DESS planning and resilience evaluation method.

2. Evaluation Framework for Highly Resilient Active Distribution Networks

Distribution networks serve as the essential “last mile” of power systems, crucial for guaranteeing a secure, dependable, and efficient electricity supply [24,25]. The rising prevalence of DG and renewable energy sources has led to significant alterations in modern distribution networks, especially in highly resilient active distribution networks, affecting both physical and operational paradigms [26,27]. These sophisticated systems must manage not only traditional load needs but also confront rising difficulties such as operational uncertainty, renewable energy intermittency, and the necessity for swift restoration. This evolution requires a thorough assessment approach that includes three fundamental dimensions: (1) Security—preserving system stability through impact resistance and self-healing features for rapid power restoration; (2) Reliability and Power Quality—guaranteeing voltage stability while reducing power fluctuations to improve supply quality; and (3) Efficiency—maximising energy storage utilisation, distributed generation integration, and dispatch coordination to enhance overall system performance.

In evaluating energy storage allocation requirements for highly resilient distribution networks, this study proposes a priority-index-based assessment framework guided by the following principles:

(1) Comprehensive demand consideration: The framework holistically addresses grid security, reliability, power quality, and efficiency, incorporating both absolute and relative metrics as well as actual versus projected values to meet complex operational demands.

(2) Systematic yet concise methodology: Unlike traditional bulk power system evaluation approaches, this study focuses specifically on active distribution networks with high resilience requirements. It establishes a streamlined indicator system for demand analysis and priority ranking to guide rational energy storage allocation.

(3) demand-driven prioritization: Storage resources are allocated based on zonal priority indices, where areas with stronger demand receive higher rankings. This ensures flexible, needs-based configuration of resilient distribution networks.

This approach provides a scientific foundation for strategic planning and energy storage allocation in highly resilient active distribution networks, ensuring stable and reliable operation. Based on the overall methodological framework shown in Figure 1, this section further details the DESS-based resilience enhancement planning process. Figure 2 illustrates the internal structure of the planning framework, including objective functions, decision variables, and the sequential planning mechanism across grid, block, and node dimensions. To further clarify the construction and application of the priority index, Figure 3 provides a step-by-step illustration of the priority evaluation and decision-guidance process.

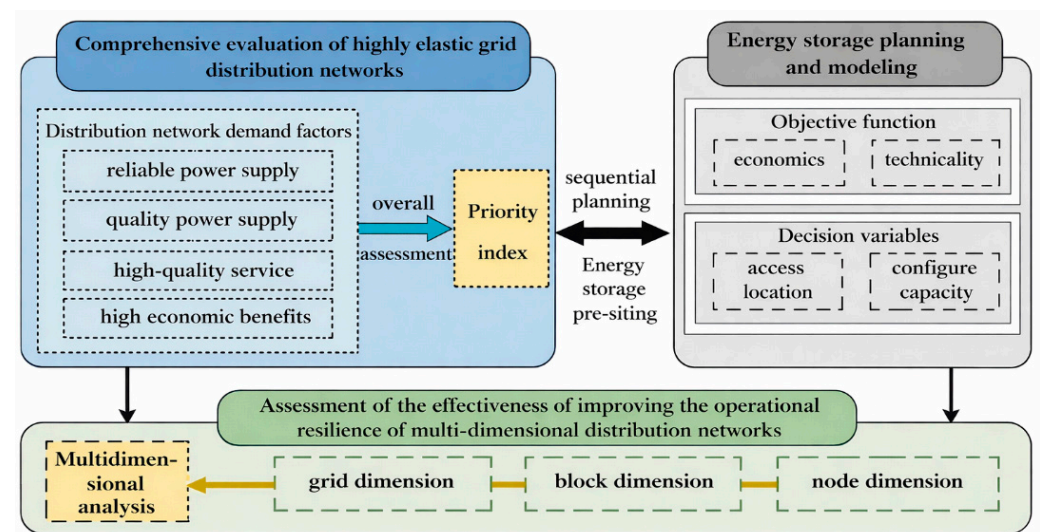


Figure 2. DESS-based resilience enhancement planning framework for distribution networks.

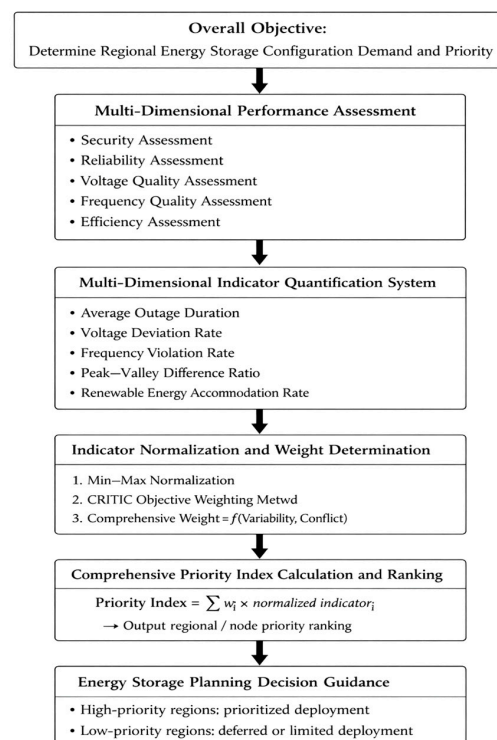


Figure 3. Block diagram illustrating the proposed overall evaluation framework.

3. Methods for Grid Resilience Enhancement Using Distributed Energy Storage

DESS can significantly enhance the resilient operation of the distribution network. In terms of emergency response, it provides backup power to critical loads, ensuring supply reliability during emergencies. For self-healing, it quickly adjusts frequency and voltage, reducing fault recovery time. In load regulation, it smooths power fluctuations by utilizing peak shaving and valley filling, strengthening grid stability. Economically, it lowers operational costs and increases profits through charging and discharging strategies. This paper combines these functions with demand evaluation, establishing a set of indicators reflecting high-quality and efficient standards, and determining the weight of each indicator through a comprehensive evaluation method. The weighted sum approach is then used to create a priority index for assessing user energy storage configuration needs. The logical structure of the proposed resilience enhancement method is outlined in Figure 4, which presents a hierarchical overview of the approaches and indicator system.

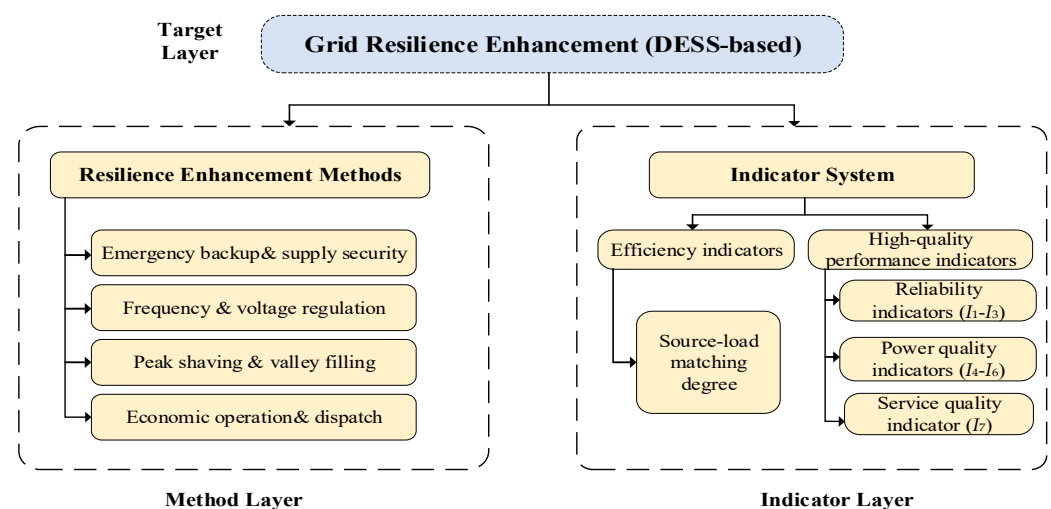


Figure 4. Hierarchical structure of DESS methods and indicators for grid resilience.

3.1. Construction of a Set of Demand Indicators

3.1.1. Efficiency Indicators

Efficiency requirements reflect the core objective of distribution networks in optimizing resource allocation and improving operational performance. With enhanced resilience, distribution networks can flexibly adapt to varying power demands and operating conditions to maximize operational efficiency. Specifically, resilient distribution networks optimize power flow through intelligent dispatch systems to maintain supply-demand balance and avoid resource waste. By implementing efficient energy management strategies like “store when cheap, discharge when needed”, they reduce storage costs while enabling rapid response to peak loads or emergencies. In renewable energy integration, these networks flexibly adjust load demand and distributed generation connections based on generalized load profiles to maximize renewable utilization and reduce dependence on upstream grids.

Generalized load refers to the aggregated net power curve formed by multiple load-related entities under normal operating conditions, including traditional demand, price-based and incentive-based demand response resources, renewable generation outputs, and energy storage operation, as illustrated in Figure 5. In this study, the generalized load concept follows the widely adopted definition in high renewable penetration systems [28], and is mainly used to characterize planning- and evaluation-level source-load interaction, rather than detailed real-time operational control. Energy storage helps mitigate renewable fluctuations and enhance regional hosting capacity, strengthening transmission coordina-

tion for self-organizing grid operation. The combination of regional load flexibility and storage improves renewable integration while maintaining stability.

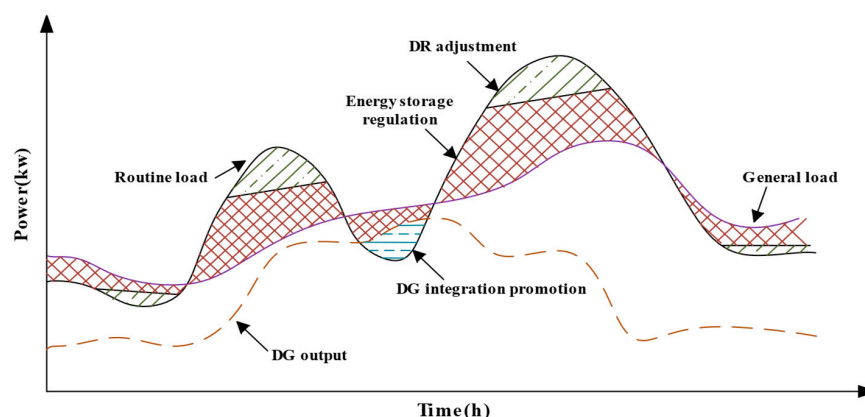


Figure 5. Generalized load curve characteristic diagram.

The generalized load retrieval model is defined as follows:

$$P_t^{\text{EL}} = P_t^{\text{base}} + P_t^{\text{DR}} + P_t^{\text{ess}} \quad (1)$$

where P_t^{EL} is the generalized load at time t considering multiple influencing factors; P_t^{base} is the conventional load demand at time t ; P_t^{DR} is the aggregated controllable load participating in demand response programs, which mainly reflects peak-shaving and load-shifting capabilities under price-based or incentive-based mechanisms, rather than emergency or interruptible load shedding; P_t^{ess} represents the energy storage charging and discharging power at time t .

The efficiency requirements of the supply area are evaluated using a matching degree metric. In the absence of energy storage systems and with demand response as the only flexibility resource, the total system load is lower than the conventional load due to the presence of reducible demand. Accordingly, this paper refines the definition of distribution network matching introduced in [29] as follows:

$$\varphi = \begin{cases} \frac{\int_0^T |P_t^{\text{EL}}| dt}{\int_0^T |P_t^{\text{DG}}| dt} \times 100\% & \int_0^T |P_t^{\text{EL}}| dt > \int_0^T |P_t^{\text{DG}}| dt \\ -\frac{\int_0^T |P_t^{\text{base}}| dt}{\int_0^T |P_t^{\text{DG}}| dt} \times 100\% & \int_0^T |P_t^{\text{EL}}| dt < \int_0^T |P_t^{\text{DG}}| dt \end{cases} \quad (2)$$

where φ is the distribution network type matching degree, used to measure the degree of source-load matching; P_t^{DG} is the new energy output at time t ; T is the number of time slots for the operation scenario.

When $\int_0^T |P_t^{\text{EL}}| dt$ is greater than $\int_0^T |P_t^{\text{DG}}| dt$, at that time, there is $\varphi \in [1, +\infty]$. A higher φ value indicates poorer DG-load matching, leading to greater grid dependence, elevated electricity procurement pressures, and increased supply-demand uncertainty. Consequently, regions exhibiting these load characteristics are identified as having high resilience enhancement requirements. When $\int_0^T |P_t^{\text{EL}}| dt$ is less than $\int_0^T |P_t^{\text{DG}}| dt$, it can be divided into two categories based on the matching degree φ . When $\varphi \in [-1, 0]$, the closer this value is to 0, the weaker the load's DG accommodation capacity becomes, indicating higher wind/solar curtailment risks and insufficient system coordination, thus defining such power-supply-characteristic regions as having high resilience enhancement potential through energy storage-enabled source-load interaction; When $\varphi \in [-\infty, 1]$, indicating the system possesses strong self-adaptive and flexible regulation capabilities, the resilience enhancement requirements are consequently reduced.

3.1.2. High-Quality Performance Indicators

High-quality requirements can be divided into three main parts: power supply reliability requirements, electrical energy quality requirements, and high-quality service requirements [30,31]. This paper further categorizes them into parameter indicators as shown in Table 2. Energy storage systems enhance distribution network resilience through a tripartite synergy: (1) For power supply reliability, they act as backup power sources to maintain critical load supply and strengthen emergency response; (2) In power quality management, they provide fast voltage/frequency regulation to boost system self-healing capacity; (3) For service quality optimization, they enable user-perception-based adaptive operation. This integrated “reliability-quality-service” mechanism empowers DESS to systematically improve grid resilience across three dimensions—emergency response, rapid restoration, and dynamic adaptation—resulting in comprehensive operational resilience enhancement.

Table 2. System of quality demand indicators.

Quality Demand Indicators	Indicator Name	Meaning of Indicators
Reliability of power supply needs	I_1 : Expectation of power supply reliability	The difference between the actual power supply reliability and the expected reliability within the statistical time
	I_2 : Primary load share	During the statistical period, the proportion of primary load in the supply area to the total load.
	I_3 : Value of loss per unit of electricity shortage	During the statistical period, the ratio of GDP to total electricity consumption in the supply area
Electricity quality needs	I_4 : Frequency nonconformance rate	The ratio of time when the power supply frequency exceeds the allowable range to the total statistical time.
	I_5 : Voltage deviation	The ratio of actual voltage to rated voltage within the specified time period.
	I_6 : Peak-to-valley difference	The ratio of the peak-to-valley difference to the maximum load over the statistical time period
Quality service needs	I_7 : Number of complaints	Total number of reliability-related customer complaints in the supply area during the statistical period, including outage complaints, power quality complaints

All indicators in this study are calculated at the planning stage using standard distribution network operation and planning data. The dataset consists of two components: typical daily nodal load profiles and annual statistical performance indicators.

The typical daily load profile of each node contains 96 sampling points with a 15 min resolution. These profiles are obtained from long-term historical measurements and are used to characterize daily load variations at each node. Together with renewable generation scenarios produced by the GMM model, these data are used to evaluate renewable–load matching and peak–valley mitigation performance, which form the basis of the efficiency indicator. Annual statistical indicators are used to quantify supply reliability, power quality, and service performance, including expected supply reliability, outage duration, frequency noncompliance, voltage deviation, and customer complaint rates. These indicators are derived from routine utility operational statistics and service records and reflect long-term operating conditions at each node. Socioeconomic statistics and electricity consumption data are used only for the interruption-cost-related indicator. Unless otherwise specified, indicators I_1 – I_7 are calculated over an annual statistical period.

This representation combines aggregated annual indicators with typical daily operating patterns, which is suitable for planning-stage analysis and avoids dependence on high-resolution full-year time-series simulations. When node-level statistics are unavail-

able, feeder- or area-level indicators are allocated to individual nodes in proportion to their load shares or customer numbers, following common planning practice.

3.2. Calculation Extraction of Typical Scenic Output Scenes Based on Improved GMM Algorithm

(a) GMM clustering model

This study employs an improved GMM algorithm to cluster representative renewable generation scenarios, aiming to characterize the uncertainty of DG outputs while effectively preserving certain extreme operating scenarios. The GMM is a probabilistic modeling approach that can effectively capture the multimodal distributions of wind and photovoltaic power outputs, which typically exhibit strong randomness as well as temporal and spatial variability. By fitting the data with multiple Gaussian components, the GMM is able to automatically identify and distinguish different uncertain generation patterns. The GMM algorithm assumes that the observed data are generated from a mixture of several Gaussian distributions [32], as expressed below:

$$P(x) = \sum_{k=1}^K \pi_k p_k(x) \quad (3)$$

$$p_k(x) = \frac{1}{(2\pi)^{D/2} |\Sigma_k|^{1/2}} \exp\left(-\frac{(x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k)}{2}\right) \quad (4)$$

where x is the set of wind and solar data samples; K is the total number of clusters in the multidimensional Gaussian distribution; D is the dimension of the data; $P(x)$ and $p_k(x)$ are the probability densities of the multidimensional GMM and the k -th Gaussian cluster; π_k and Σ_k and μ_k are the coefficients, covariance matrices, and mean vectors of the k -th Gaussian cluster, respectively.

The GMM algorithm achieves optimal parameter estimation by maximizing the log-likelihood function, as shown below:

$$\max \sum_{n=1}^N \log\left(\sum_{k=1}^K \pi_k p_k(x_n)\right) \quad (5)$$

where N is the total number of sample groups; x_n is the n th group of scenery samples; π_k satisfies $0 < \pi_k \leq 1$ and $\sum_{k=1}^K \pi_k = 1$. The optimization problem is frequently addressed through the utilization of the expectation-maximization algorithm.

(b) Parameter initialisation

The key to solving the GMM algorithm lies in determining the initial values of the sub-Gaussian models. This article uses the K-means clustering method based on the RV coefficient [33] to initialize the Gaussian distribution parameters.

The RV coefficient, as a statistical measure of the degree of linear relationship in a matrix, is applied to measure the distance between the sample matrix X_n and cluster centers. By iteratively updating the cluster centers, the matrix X can be partitioned into K clusters. The formula for the RV coefficient is as follows:

$$RV(X_n, C_m) = \frac{\text{tr}(X_n^T C_m)}{\sqrt{\text{tr}(X_n^T X_n) \text{tr}(C_m^T C_m)}} \quad (6)$$

where X_n is the n th sample matrix; C_m is the k th cluster center matrix; $\text{tr}(\cdot)$ is the trace of the matrix.

(c) Optimal number of clusters determined

The optimal clustering outcome necessitates maximizing the similarity among sample points within the same class, emphasizing minimal intra-class distances. Simultaneously, it requires maximizing the dissimilarity between sample points belonging to different classes, emphasizing larger inter-class distances. This principle underscores the dual optimization goals of cluster analysis, focusing on enhancing within-class homogeneity and promoting between-class heterogeneity. Therefore, the evaluation function for clustering effectiveness is defined as follows [34]:

$$CH(K) = \frac{D_{\text{out}}(K)/(K-1)}{D_{\text{in}}(K)/(N-K)} \quad (7)$$

$$D_{\text{in}}(K) = \sum_{k=1}^K \sum_{x \in C_k} \|x - \mu_k\|^2 \quad (8)$$

$$D_{\text{out}}(K) = \sum_{k=1}^K |C_k| \cdot \|\mu_k - \mu\|^2 \quad (9)$$

where $D_{\text{in}}(K)$ is the intra-class sample distance; $D_{\text{out}}(K)$ is the aggregate of sample distances among categories; μ is the global mean of all samples; C_k is the k th cluster; $|C_k|$ is the number of samples in the cluster. When $CH(K)$ is at its peak, the associated K represents the optimal number of clusters.

3.3. Construction of Priority Indices

The priority index measures the intensity of energy storage demand in distribution networks, based on a multi-criteria assessment of demand indicators. The methodology for determining the priority index is articulated as follows:

$$H_{\text{pi}} = \sum_{a=1}^7 \omega_a I'_a + \phi' \quad (10)$$

where H_{pi} represents the priority index for supply areas, indicating that a higher priority index corresponds to a greater demand for energy storage configuration. Consequently, it should be given more consideration in DESS planning site selection; ω_a ($a = 1, 2, \dots, 7$) is the weight of the quality index, which totals 1; I'_a and ϕ' stand for the dimensionless quality and efficiency indexes after normalization.

The Critic approach, an objective weighting technique, allocates weights in a multi-indicator comprehensive evaluation by analysing the correlation and disparities among indicators, thus scientifically assessing the significance of each indicator [35]. The procedure is as follows:

(1) Dimensionless processing

Conducting dimensionless on a set of b samples with p high-quality performance indicators to avoid the impact on evaluation caused by varying value ranges and dimensions.

$$\begin{cases} x'_{d,c+} = \frac{x_{d,c} - \min(x_d)}{\max(x_d) - \min(x_d)} \\ x'_{d,c-} = \frac{\max(x_d) - x_{d,c}}{\max(x_d) - \min(x_d)} \end{cases} \quad (11)$$

where $x_{d,c}$ and $x'_{d,c}$ are the raw and dimensionless values of the quality indicator for item d of the c th sample, respectively; $\max(x_d)$ and $\min(x_d)$ are the maximum and minimum values of the index d in all samples, respectively. In this paper, desired supply reliability is a reverse indicator, while the others are positive indicators.

(2) Assessment of internal variability and interdependencies in quality indicators

The standard deviation quantifies the extent of internal variation in quality measures. The Pearson correlation coefficient indicates the extent of association between two quality metrics.

$$S_d = \sqrt{\frac{1}{b} \sum_{c=1}^b (x'_{d,c} - \bar{x}'_d)^2} \quad (12)$$

$$r_{de} = \frac{\sum_{c=1}^b (x'_{d,c} - \bar{x}'_d)(x'_{e,c} - \bar{x}'_e)}{\sqrt{\sum_{c=1}^b (x'_{d,c} - \bar{x}'_d)^2 \sum_{c=1}^b (x'_{e,c} - \bar{x}'_e)^2}} \quad (13)$$

where S_d is the standard deviation of the d th quality indicator, the closer S_d is to 1 the greater the variability within the indicator. r_{de} is the correlation coefficient between the d th and e th quality indicators; $\bar{x}'_d$ is the mean value of the d th evaluation indicator.

(3) Obtain objective weights for each indicator

The weights are determined based on the magnitude of information provided by various quality indicators.

$$\begin{cases} M_d = S_d \sum_{e=1}^p (1 - r_{de}) \\ \omega_d = \frac{M_d}{\sum_{d=1}^p M_d} \end{cases} \quad (14)$$

where M_d is a measure of informativeness, with larger values indicating that the quality indicator d contains more information. The quality indication d becomes heavier as the amount of information increases.

3.4. Case Studies

3.4.1. Basic Overview

This study employs an actual power supply grid as a case study. In engineering practice, block partitioning is often guided by internal grid-planning guidelines. In this study, the partitioning results are adopted from existing planning outcomes and are consistent with publicly available standards such as DL/T 5729-2023 [36]. Zhejiang Province is selected as the case study region due to its representativeness of highly resilient active distribution networks in China. The province features a high penetration of distributed photovoltaic and wind generation, together with diversified load types including residential, industrial, commercial, and public service sectors, resulting in complex and heterogeneous source–load characteristics. In addition, Zhejiang has well-established grid-based planning practices and comprehensive historical datasets for renewable generation and load categorization, which provide a solid foundation for priority index construction and scenario-based planning. The coexistence of load-dominant and generation-dominant blocks within the same distribution network further makes Zhejiang an appropriate testbed for evaluating priority-guided and sequential DESS planning strategies under realistic operating conditions.

Block partitioning is performed by comprehensively considering feeder topology, geographical proximity, and end-user functional similarity. Each block aggregates loads with similar consumption characteristics and service attributes, ensuring internal homogeneity while preserving clear electrical and spatial boundaries between blocks. As a result, six types of end-user blocks are identified, including residential, industrial, commercial, administrative, educational, and medical sectors, which collectively represent nine typical load profiles. The specific block division results and the corresponding load curve characteristics are illustrated in Figures 6 and 7, respectively.

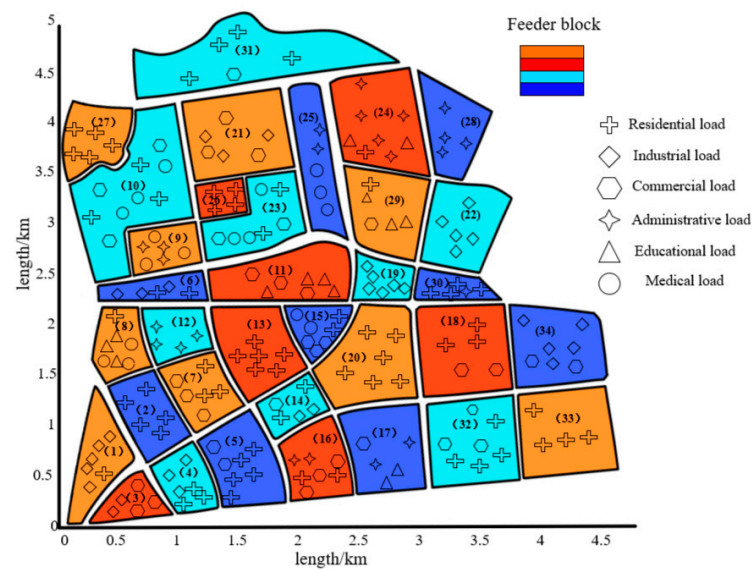


Figure 6. Grid block distribution schematic.

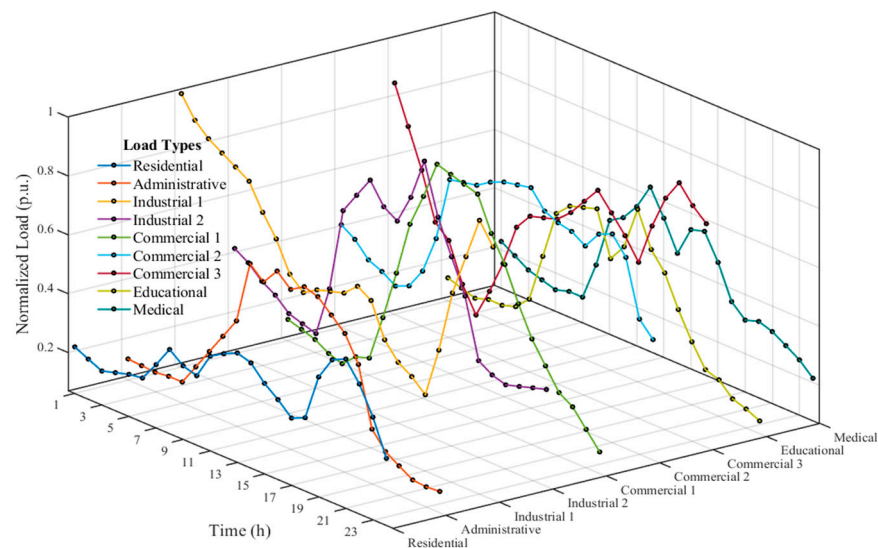


Figure 7. Load typical daily output curve.

China implements a time-of-use electricity pricing system with differentiated rates for residential and industrial/other consumers. Residential users face peak rates of 0.563 yuan/kWh from 8:00 to 22:00 and off-peak rates of 0.291 yuan/kWh from 22:00 to 8:00. For industrial and commercial consumers, pricing follows a three-tier structure: peak rates (1.092 yuan/kWh) apply during 19:00–21:00, standard rates (0.925 yuan/kWh) during 8:00–11:00, 13:00–19:00, and 21:00–22:00, while off-peak rates (0.412 yuan/kWh) are charged from 22:00 to 8:00 and 11:00–13:00. This pricing mechanism incentivizes load shifting to balance grid demand and optimize power resource utilization. The reducible load capacity is configured to 10–20% of the nodal load, excluding critical demand sectors such as administrative, medical, and educational facilities from participating in demand response programs [37]. The model assumes uniform power output characteristics among all DG units while considering variations in their installed capacities. DG deployment is restricted to industrial, commercial, and residential load nodes, with a surplus power feed-in tariff rate of 0.4153 yuan/kWh.

3.4.2. Typical Scenario Generation for Wind and Solar Power Output

The improved GMM algorithm was applied to extract typical grid scenes of grid landscapes as described earlier. The calculation results indicate that the clustering validity evaluation function is maximized when the number of scenes is 5. The corresponding curves of typical grid scenes and the probabilities of each scene can be found in Figures 8 and 9.

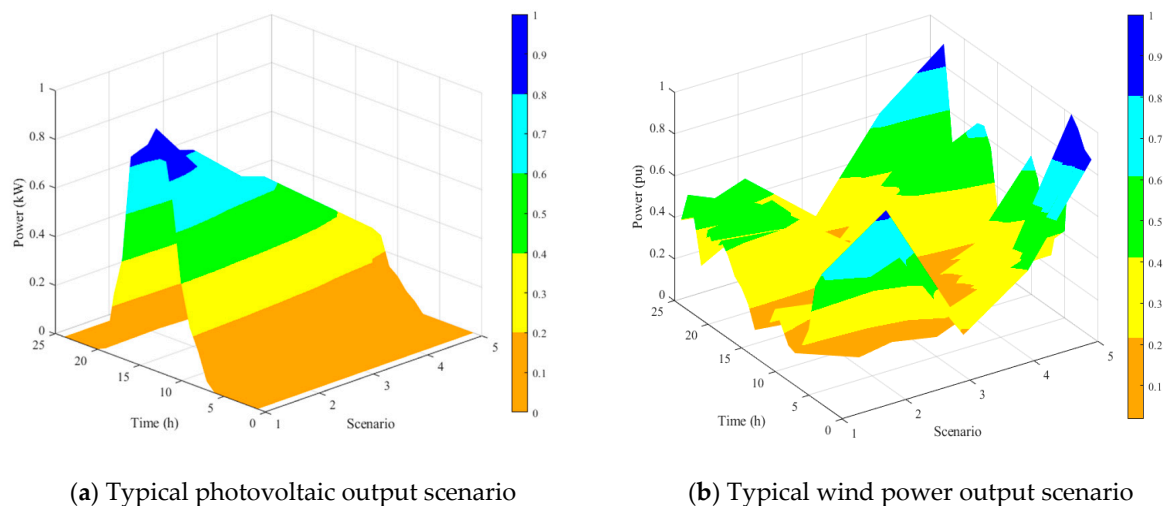


Figure 8. Representative scenarios of wind and solar power output.

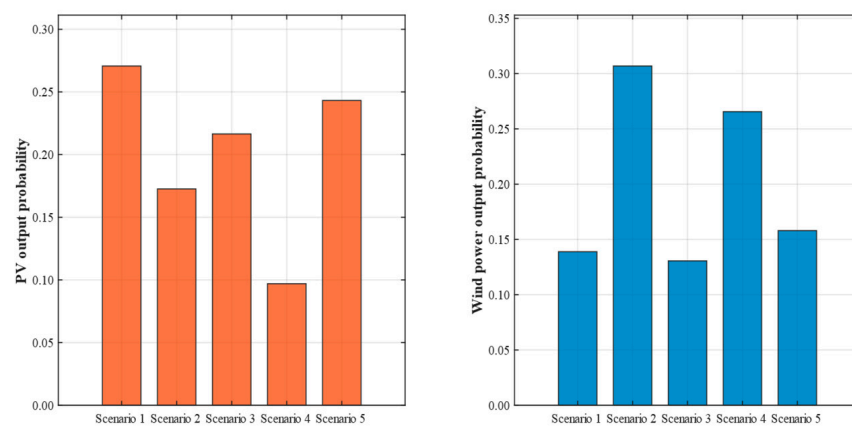


Figure 9. Probability of typical power output scenarios.

The results demonstrate that the typical output scenarios of photovoltaic and wind power systems, extracted using the improved GMM algorithm, display distinct clustering characteristics in their three-dimensional spatiotemporal distribution. PV output scenarios exhibit clear diurnal patterns with active generation during daylight hours and complete shutdown at night, while maintaining well-defined output gradient distributions across different scenarios. In contrast, wind power output scenarios present more complex multimodal characteristics, showing significant variations in both temporal distribution and output patterns among different scenarios.

The hierarchical differentiation observed in PV output scenarios and the diverse fluctuation patterns of wind power scenarios collectively validate the algorithm's effectiveness in capturing the essential operational characteristics of distributed generation. By applying scenario probabilities as weighting factors, we can derive a comprehensive weighted grid-based integrated output curve for PV and wind power systems.

3.4.3. Priority Index Construction Results

Through the proposed Critic method, the weights of the quality indicators I_1 to I_7 are obtained as 0.05, 0.19, 0.15, 0.08, 0.18, 0.17, and 0.19, respectively.

These indicators demonstrate the distribution network's flexibility and dependability in response to atypical events such as load fluctuations and equipment malfunctions. The Critic technique assesses indicator weights, elucidating their relative significance for network resilience. The primary load ratio and unit outage loss value are crucial, emphasising the importance of essential supply and the reduction in outage costs. Voltage fluctuations, peak-to-valley discrepancies, and complaint frequencies further emphasise system stability and service quality. These indicators proficiently direct network optimisation, guaranteeing enough resistance against adversities.

3.4.4. Matching Results for Each Block

Figure 10 illustrates the source-load matching degree and characteristics for each grid block based on the aggregated wind and solar power output. The analysis reveals significant spatial variations in source-load characteristics across different grid blocks.

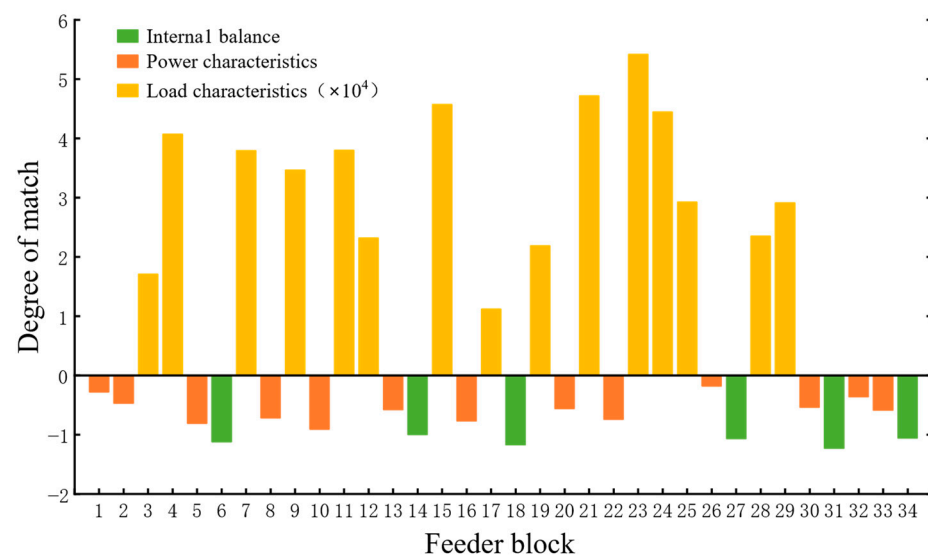


Figure 10. Identification of matches and types of blocks of the grid.

Notably, blocks 21 and 23 demonstrate distinct load-dominant characteristics with no local DG output, while blocks 1 and 26 exhibit pronounced generation-dominant features, where DG power supply exceeds local consumption by more than threefold. These blocks with extreme source-load imbalance should be prioritized in the DESS siting process to optimize grid performance.

4. Multi-Objective Energy Storage Planning Based on Sequential Optimisation

The current centralized planning approach for DESS sometimes results in local over-sizing configurations. This paper presents a sequential planning strategy for DESS to attain global optimization of solutions by enhancing the planning process. Initially, it is essential to acquire DG representative output scenarios that reflect several locations to delineate their operational uncertainty. A DESS capacity allocation model is subsequently developed by thoroughly evaluating technical and economic aspects.

4.1. DESS Sequential Planning Solution Process

The process of DESS sequential planning based on priority index is shown in Figure 11. In the proposed sequential planning framework, DESS installation locations are determined by the priority index ranking. At each iteration, the node with the highest priority index is selected as the candidate site, and the optimization model is then applied to determine the corresponding storage capacity and operating schedule at that node. After the DESS is deployed, system indicators and generalized load curves are updated, and the priority index is recalculated to guide the next iteration of siting and sizing until the budget constraint is satisfied.

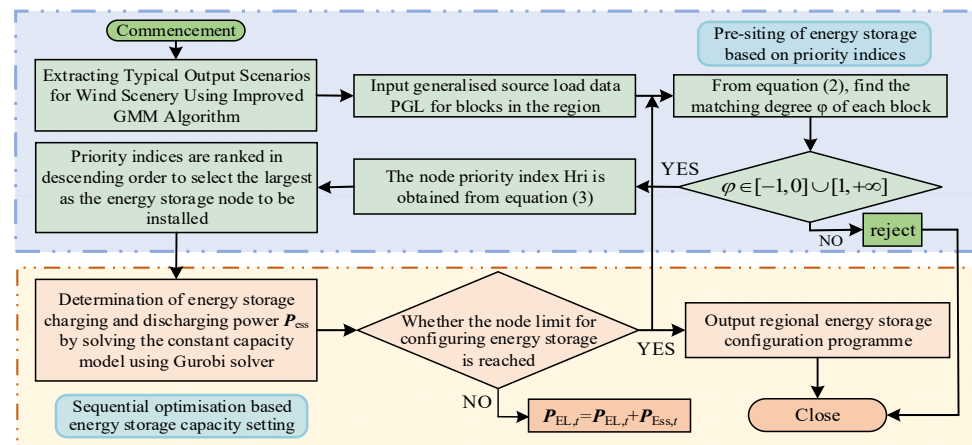


Figure 11. Planning process diagram.

- (1) Based on historical data of wind speed and light intensity within the grid, utilize the enhanced GMM algorithm to extract typical wind and solar power output scenarios.
- (2) Analyze the load data of each grid block, aggregate the demand response capabilities of each block to derive the corresponding generalized load curve.
- (3) Calculate the matching degree of each block according to Formula (2), and select blocks with load characteristics or power source characteristics as the pre-selected blocks for installing DESS.
- (4) Read the quality parameters and efficiency parameters of each node within the block, calculate the priority index of each node according to Formula (3);
- (5) Find the node with the highest priority index and use it as the DESS configuration node.
- (6) Solve the multi-objective optimization configuration model of DESS to obtain the corresponding DESS capacity and charge–discharge power.
- (7) Based on the DESS settings, update the generalized load curve for the block including the node.
- (8) Repeat steps 3–7 above, while retaining the previous DESS configuration, until the cumulative equivalent annual investment cost reaches the predefined budget limit.

The proposed DESS planning framework targets a predefined planning horizon and investment scale specified by the utility, reflecting practical distribution-level decision-making under uncertainty rather than long-term capacity expansion planning.

4.2. DESS Multi-Objective Capacity Allocation Model

In each iteration, the optimization model determines the sizing and operational decisions of the DESS at the selected node. The decision variables include the rated energy capacity and rated power of the DESS, the charging and discharging power, and the state of charge at each time step t under renewable generation scenario s . Binary variables are introduced to represent mutually exclusive charging and discharging states. The instal-

lation location is fixed in each iteration based on the priority index ranking described in Section 3 and is not treated as an optimization variable within the capacity allocation model. Accordingly, the optimization jointly determines the DESS capacity configuration and its time-series operating profile under system-level constraints.

4.2.1. Objective Function

To enhance the absorption capacity of DESS for renewable energy integration, facilitate source-load matching, and enhance power supply reliability, a DESS capacity allocation model is formulated. This model aims to optimize the operation cost F_1 , economic benefit F_2 , wind and solar curtailment rate F_3 , and the and peak-to-valley difference F_4 throughout the DESS lifecycle, as shown in Formula (15) to (22).

$$F_1 = \text{CRF} \cdot (C_{\text{ess}}^1 + C_{\text{ess}}^0) \quad (15)$$

$$\text{CRF} = \frac{\beta(1+\beta)^Y}{(1+\beta)^Y - 1} \quad (16)$$

$$C_{\text{ess}}^1 = P_{\text{ess}} \cdot c_p + E_{\text{ess}} \cdot c_e \quad (17)$$

$$C_{\text{ess}}^0 = \sum_{o \in Y} \frac{(P_{\text{ess}} \cdot c_{\text{op}} + E_{\text{ess}} \cdot c_{\text{oe}})}{(1+\beta)^o} \quad (18)$$

$$F_2 = B_{\text{sale}} + B_{\text{save}} \quad (19)$$

$$F_3 = \frac{\sum_{t \in T} (P_t^{\text{DG,max}} - P_t^{\text{DG}}) \cdot \Delta t}{\sum_{t \in T} P_t^{\text{DG}}} \quad (20)$$

$$F_4 = \frac{P_{\text{max}} - P_{\text{min}}}{P_{\text{max}}} \quad (21)$$

$$\min(F_1, F_2, F_3, F_4) \quad (22)$$

where β is the discount rate, which is taken to be 0.05 in this paper; Y is the working life of DESS; C_{ess}^1 is the one-time construction cost of energy storage; C_{ess}^0 is the present value of the operation and maintenance costs of energy storage over its lifecycle; B_{sale} is the present value of the revenue from electricity sales obtained by the node configured with energy storage; B_{save} is the present value of the cost of electricity purchased from the higher-level grid saved by the node configured with energy storage; P_{ess} and E_{ess} are the power and capacity of energy storage, respectively; c_p and c_e are the unit installation cost of storage power and capacity, respectively; c_{op} and c_{oe} are the unit O&M cost of DESS power and capacity, respectively; Δt is the duration of a single time period of an operation scenario; $P_t^{\text{DG,max}}$ and P_t^{DG} are the scenery unit's maximum and actual output at moment t ; P_{max} and P_{min} are the maximum and minimum loads of the node.

All objective functions are evaluated based on the time-series operating results obtained from the optimization model under all renewable generation scenarios. Economic cost and benefit objectives are calculated by aggregating investment-related terms and operation-related revenues over the DESS lifecycle. Technical performance objectives are derived from time-series system responses, including total renewable energy curtailment and peak–valley difference of the generalized load curve after DESS scheduling. These aggregated indicators are used to characterize long-term economic and operational performance at the planning stage. To solve the multi-objective optimization problem, all objective functions are first normalized with respect to their corresponding base-case values and then combined into a single composite objective using a weighted-sum method. The resulting mixed-integer nonlinear programming problem is solved using the Gurobi optimizer.

The proposed multi-objective DESS planning model integrates four complementary objectives, including lifecycle cost minimization, operational benefit maximization, renewable energy curtailment minimization, and peak-to-valley load variance reduction. Objectives F_1 and F_2 jointly characterize the economic feasibility of DESS deployment by coordinating investment affordability and operational profitability, while F_3 and F_4 focus on technical performance by enhancing renewable energy accommodation and load smoothing capability. Rather than being optimized independently, these objectives are incorporated into a unified optimization framework and coordinated through the sequential siting and sizing strategy described in Section 4.1.

At each iteration, the optimization process seeks a balanced compromise among economic and technical objectives under system constraints, resulting in a coordinated DESS configuration that improves grid resilience without sacrificing economic viability.

4.2.2. Constraint Conditions

The constraints ensure the physical feasibility, operational consistency, and planning-level deployment consistency of the DESS, including system-level power balance, charging and discharging state limits, charging and discharging power limits, energy storage energy balance, DESS capacity bounds, SOC constraints, and an overall planning budget constraint. These constraints are detailed as follows:

(1) Constraints on charging and discharging power limits.

$$P_t^{\text{ess,min}} \leq P_t^{\text{ess}} \leq P_t^{\text{ess,max}} \quad (23)$$

where $P_t^{\text{ess,max}}$, $P_t^{\text{ess,min}}$ are the upper and lower limits of the charging and discharging power of the stored energy at time t , respectively.

(2) Charge and discharge state constraints.

$$\mu_t^{\text{ess,c}} + \mu_t^{\text{ess,dis}} \leq 1 \quad (24)$$

where $\mu_t^{\text{ess,c}}$ is the state of charge of the energy storage at time t , charging is 1 and discharging is 0; $\mu_t^{\text{ess,dis}}$ is the state of discharge of the energy storage at time t , discharging is 1 and charging is 0. At the same time, energy storage can only be in either charging or discharging state.

(3) System-level power balance constraint.

$$P_{t,s}^{\text{grid}} + P_{t,s}^{\text{DG}} = P_{t,s}^{\text{EL}} + P_t^{\text{ess}}, \forall t, s \quad (25)$$

where $P_{t,s}^{\text{grid}}$ is the power exchanged with the upstream grid under scenario s , $P_{t,s}^{\text{DG}}$ is the renewable generation output, $P_{t,s}^{\text{EL}}$ is the load demand, and P_t^{ess} is the charging/discharging power of the DESS.

(4) Energy storage capacity balance constraint.

$$\sum_{t \in T} (\eta_{\text{ess}}^{\text{c}} \cdot P_t^{\text{ess}} \cdot \mu_t^{\text{ess,c}} \cdot \Delta t + \frac{1}{\eta_{\text{ess}}^{\text{dis}}} P_t^{\text{ess}} \cdot \mu_t^{\text{ess,dis}} \cdot \Delta t) = 0 \quad (26)$$

where $\eta_{\text{ess}}^{\text{c}}$ and $\eta_{\text{ess}}^{\text{dis}}$ are the charging and discharging efficiencies of the energy storage.

(5) DESS capacity upper and lower bound constraints.

$$E_{\text{ess}}^{\text{min}} \leq E_{\text{ess}} \leq E_{\text{ess}}^{\text{max}} \quad (27)$$

where $E_{\text{ess}}^{\text{max}}$ and $E_{\text{ess}}^{\text{min}}$ are the upper and lower limits of the capacity of the energy storage configuration, respectively.

(6) Charge state constraint.

$$E_t^{\text{soc}} = E_{t-1}^{\text{soc}} + \mu_t^{\text{ess},c} \cdot P_t^{\text{ess}} \cdot \Delta t \cdot \eta_{\text{ess}}^c - \frac{\mu_t^{\text{ess},dis} \cdot P_t^{\text{ess}} \cdot \Delta t}{\eta_{\text{ess}}^{\text{dis}}} \quad (28)$$

$$\text{SOC}^{\min} E_{\text{ess}} \leq E_t^{\text{soc}} \leq \text{SOC}^{\max} E_{\text{ess}} \quad (29)$$

where E_t^{soc} is the amount of energy stored at time t ; $\text{SOC}^{\max}$ and $\text{SOC}^{\min}$ are the upper and lower bound constraints on energy storage, respectively.

The constructed DESS multi-objective planning model is a mixed-integer nonlinear planning problem, which is solved using the Gurobi solver.

(7) Overall planning budget constraint:

$$\sum_{m=1}^M \left(\text{CRF} \cdot C_m^{\text{inv}} + C_m^{\text{OM,year}} \right) \leq B^{\text{EAV}} \quad (30)$$

where m denotes the index of installed DESS units following the sequential planning process; M is the total number of installed DESS units (not an optimization variable); C_m^{inv} represents the one-time capital investment cost of the m -th DESS unit; $C_m^{\text{OM,year}}$ denotes the annual operation and maintenance cost of the m -th DESS unit; CRF is the capital recovery factor; and B^{EAV} is the equivalent annual investment budget specified by the utility at the planning stage.

The budget B^{EAV} is specified by the utility or system owner at the upper planning level based on engineering feasibility and financial constraints. Therefore, the number of installed DESS units is implicitly determined by the budget-constrained sequential process.

4.3. Case Settings and Results

To validate the effectiveness of the proposed priority index method for DESS sequential planning, three representative case studies are conducted. In addition, to clearly illustrate the overall system architecture and the interactions among its main components, a schematic diagram of the DESS is provided in Figure 12. The diagram depicts the key elements of the system, including wind and photovoltaic generation units, the energy storage system, and their electrical interconnections within the distribution network.

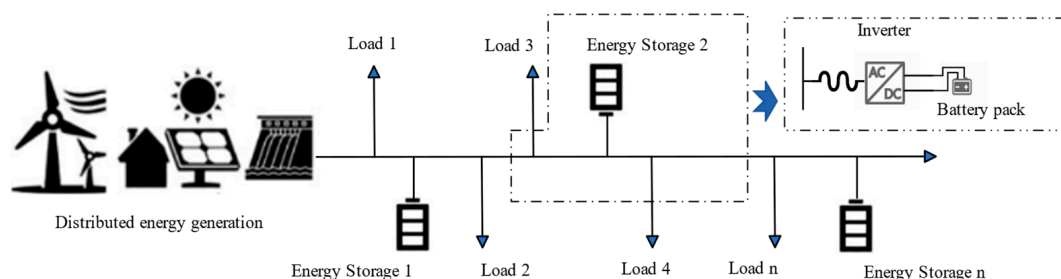


Figure 12. Schematic Diagram of DESS.

Case 1: Utilizing a global traversal method for centralized site selection. Directly planning the capacity and power to be connected to the DESS for all nodes within the grid, determining the installation nodes of DESS based on sorting the objective function values from small to large.

Case 2: Utilizing a priority index method for site selection. Calculate the priority index of all nodes within the grid, determine several DESS installation nodes based on the descending order of the priority index, and plan their required capacity and power access accordingly.

Case 3: Sequential site selection using a priority index.

Based on this system configuration, a series of representative case studies are conducted to evaluate the effectiveness of the proposed priority-index-based DESS planning strategy. To facilitate the interpretation of the obtained results, a result-oriented workflow is illustrated in Figure 13, which outlines the logical connection between priority index evaluation, sequential siting and sizing of DESS, and the subsequent assessment of system-level electrical performance and comparative case analysis.

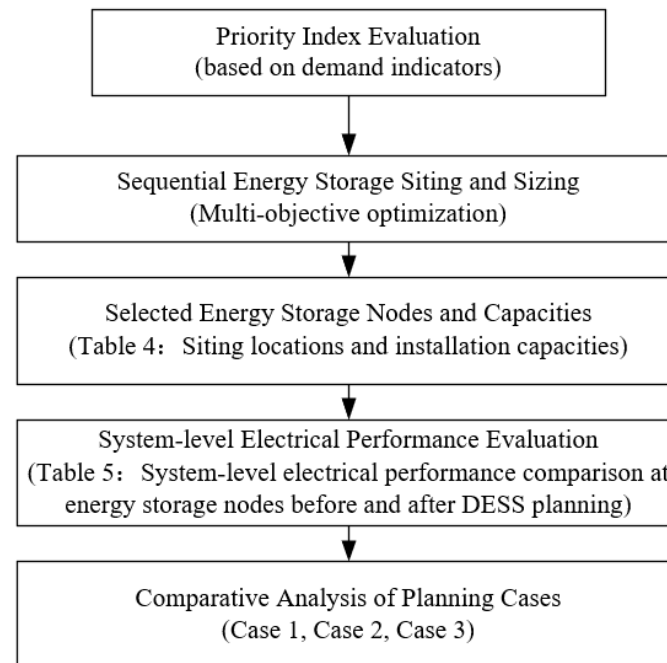


Figure 13. Result-oriented process flow diagram.

To further clarify how the results reported in this section are generated, including the inputs, computational tools, outputs, and their correspondence with the methods introduced in previous sections, a summary of the result generation process is provided in Table 3. The table explicitly maps each stage of the workflow to its associated input data, solution method, reported results, and related sections of the paper.

Table 3. Result generation process summary: inputs, solution methods/tools, outputs, and corresponding sections.

Stage	Input	Method/Tool	Output	Related Section
Demand evaluation framework	Grid security, reliability, power quality, efficiency requirements	Node–block–grid evaluation framework	Demand dimensions and indicator categories	Section 2
Priority evaluation	Demand indicators	Priority index model (Critic-based weighting)	Priority indices	Section 3
DESS planning	Priority indices	Multi-objective optimization (Python, Gurobi)	DESS locations and capacities	Section 4.2/ Table 4
Electrical evaluation	DESS configuration	Power system simulation	P–V, frequency, voltage metrics	Table 5
Case comparison	Planning results	Comparative analysis	Case-level insights	Section 4.3

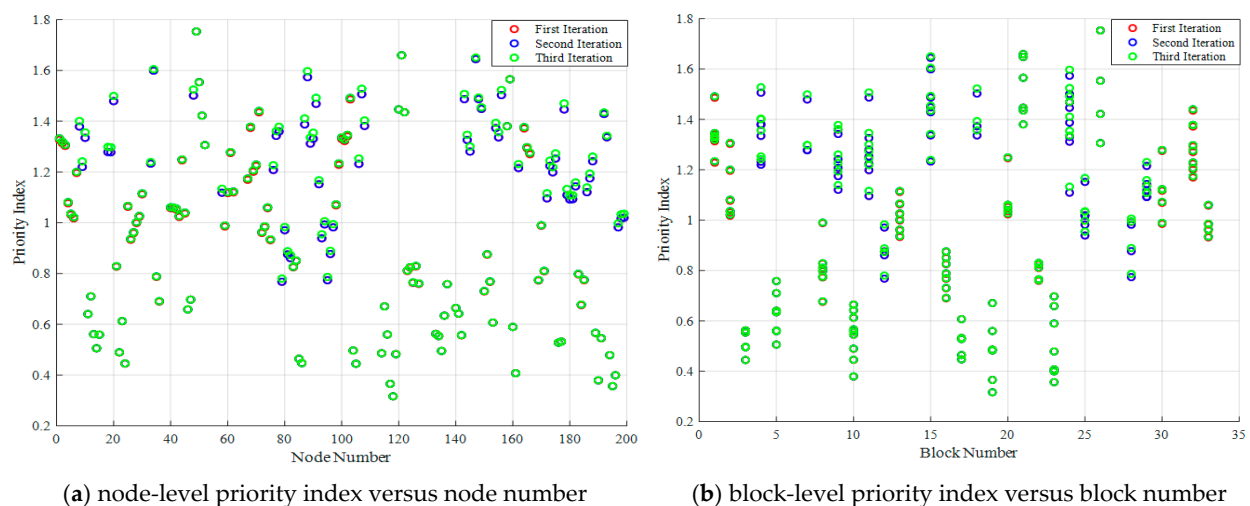
Table 4. Equipment Parameters of Each DESS.

Parameter	Numerical Value
Cost per unit capacity/(CNY·(kW)-1)	2000
Unit power cost/(CNY·(kW)-1)	600
Annual O&M cost per unit of capacity/(CNY·(kW)-1)	10
Annual O&M cost per unit of power/(CNY·(kW)-1)	90
Charge and discharge efficiency	0.95
Energy storage upper limit factor and lower limit factor	0.9, 0.2
Service life/year	12

Table 5. Corresponding storage planning results for each case.

Case	1	2	3
Average primary load share	0.33	0.60	0.62
Maintenance cost/(CNY/year)	443,400	732,600	762,600
Economic benefits/(CNY/year)	136,300	455,900	865,200
Configure energy storage nodes	20,41,70	49,121,157	49,121,147
Installation capacity/(kW·h)	632,605,498	733,1166,1076	733,1166,1176
Belonging Block	7,20,32	26,21	26,21,15

A total of three units of DESS are planned and constructed in each case, and the corresponding parameter settings are summarized in Table 4. In this case study, the utility specifies an overall investment scale corresponding to an equivalent annual budget of approximately. Under this budget constraint, the proposed sequential planning framework determines that the optimal solution consists of three DESS installations. The priority indices obtained from three iterations for all candidate nodes are presented in Figure 14. The DESS capacity planning model is implemented in Python (v3.9), and the multi-objective optimization problems are solved using the Gurobi (v11.0) optimizer, yielding the final planning cases for each method.

**Figure 14.** Priority indices obtained from each node iteration.

As illustrated in Figure 14, priority indices are first calculated at the node and block levels to quantify the relative urgency of energy storage demand. Nodes with higher priority indices are preferentially selected as candidate locations for energy storage deployment, providing guidance for the subsequent siting and sizing process.

Based on the above objective formulation, the proposed multi-objective optimization framework seeks solutions that balance economic feasibility and operational performance

under multiple constraints. The coordinated optimization of cost-related and technical objectives enables the model to simultaneously improve load balancing, operational quality, and economic outcomes. The impacts of the multi-objective planning strategy are quantitatively reflected in the results reported in Tables 5 and 6. Table 5 summarizes the storage siting and sizing outcomes under different planning cases, highlighting the economic performance and spatial allocation characteristics. Table 6 further presents the corresponding system-level electrical performance at the energy storage nodes, including peak–valley difference, frequency nonconformance rate, and voltage deviation, demonstrating the operational improvements achieved through coordinated optimization. These improvements are mainly attributed to the objective that promotes load balancing through coordinated energy storage scheduling.

Table 6. System-level electrical performance comparison at energy storage nodes before and after DESS planning.

Case	Block	Load Type	Node	Original P–V (%)	Current P–V (%)	Freq. Violation (%)	Voltage Deviation (%)
1	7	Residential	20	70.9	21.9	7.960	3.490
	20	Residential	41	75.2	24.1	1.900	7.810
	32	Residential	70	72.0	30.6	6.950	9.490
2	26	Residential	49	74.9	36.9	9.700	6.330
	21	Industrial	121	79.2	43.0	5.340	8.230
	21	Commercial	157	77.3	34.0	5.410	8.140
3	26	Residential	49	74.9	36.9	9.700	6.330
	21	Industrial	121	79.2	43.0	5.340	8.230
	15	Commercial	147	75.1	38.2	6.030	8.540

Table 5 summarizes the storage siting and sizing outcomes for the three planning cases, which directly reflect the trade-offs achieved by the proposed multi-objective optimization framework. Specifically, the selected storage nodes and their capacities represent the compromise solution among the economic cost objective, operational benefit objective, and technical performance-related objectives. Case 1 yields a relatively scattered configuration (nodes 20, 41, and 70 with capacities of 632, 605, and 498 kW·h), while Case 2 selects nodes 49, 121, and 157 (733, 1166, and 1076 kW·h) based on a one-time priority ranking and exhibits local concentration because nodes 121 and 157 both belong to Block 21. By sequentially updating the priority index, Case 3 adaptively adjusts the siting decision after each installation and finally selects nodes 49, 121, and 147 (733, 1166, and 1176 kW·h), which avoids excessive concentration and achieves a more balanced configuration. Case 3 avoids the excessive spatial concentration observed in Case 2 while achieving the highest economic return among all cases. From the economic perspective, Table 5 also shows that the planning cases differ in annualized costs and economic benefits, indicating that the proposed model can coordinate investment affordability and operational profitability within a unified optimization framework.

To assess the system-level electrical operating impact of the proposed planning and optimization strategy, Table 6 reports the corresponding electrical performance at the energy storage nodes before and after DESS planning, including peak–valley difference, frequency nonconformance rate, and voltage deviation. These indicators provide direct evidence of operational improvements achieved by the optimized storage configuration. In particular, the reduction in peak–valley difference indicates enhanced load smoothing capability, which alleviates peak stress at the node/block level and supports operational flexibility. Meanwhile, changes in frequency nonconformance rate and voltage deviation reflect improvements in regulation capability and voltage quality at the storage nodes,

which are closely related to system resilience enhancement. By jointly considering the economic outcomes in Table 5 and the electrical performance improvements in Table 6, the results demonstrate that the proposed multi-objective framework improves technical operating performance without sacrificing overall economic feasibility.

Figure 15 further illustrates the demand indicator profiles of the selected energy storage nodes under different planning cases. In Case 1, the global traversal strategy results in larger variations in demand indicator values across the selected nodes, indicating weaker correspondence to local demand characteristics and explaining the longer computation time. In contrast, Cases 2 and 3 select nodes with relatively higher demand indicator values, suggesting that the siting decisions are more aligned with actual demand intensity.

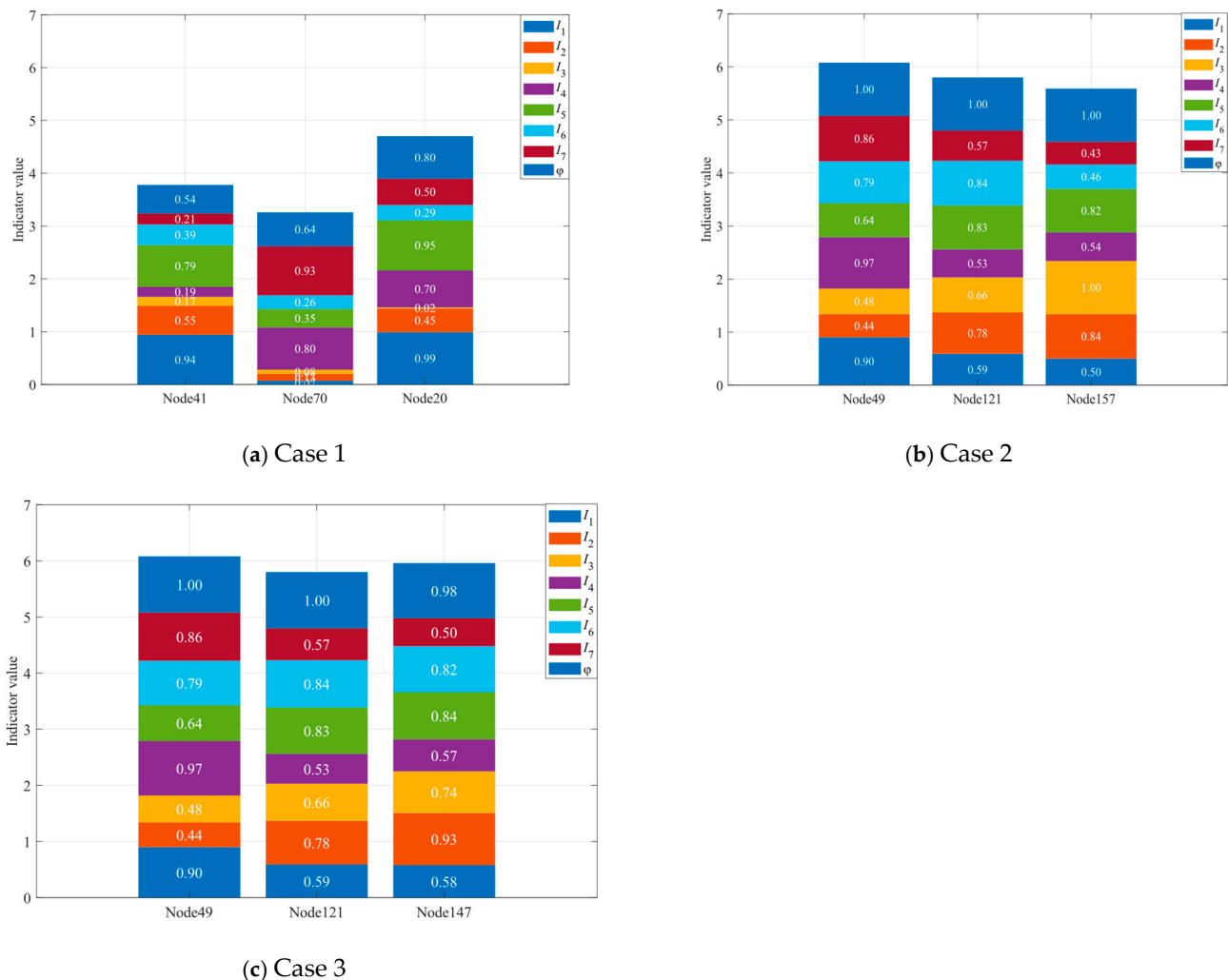


Figure 15. Demand indicator chart for selected nodes in each case.

In summary, the sequential updating strategy explains the balanced excellence of Case 3: it avoids local over-concentration without compromising demand-based prioritization.

5. Multi-Dimensional Evaluation for Distribution Network Performance Enhancement

Validation of energy storage design impacts on the self-recovery capability and stable operating performance of highly robust active distribution networks in changeable settings, using a quantitative assessment framework. During the evaluation, several metrics are proposed for a thorough assessment across three dimensions: node, block, and grid.

All indicators are evaluated based on the original physical distribution network, its feeder topology, and actual operational data. The node–block–grid structure is adopted solely as an organizational framework to support performance assessment across multiple spatial and planning scales, while maintaining full consistency with the underlying physical network model. In this framework, nodes correspond to physical buses and candidate locations for distributed DESS, blocks represent planning-oriented regional units defined according to grid-based planning standards, and the grid denotes the entire distribution network. Based on this hierarchical structure, resilience-related indicators are systematically organized and evaluated to quantify the impacts of energy storage deployment at different spatial levels. It should also be noted that all node-, block-, and grid-level indicators are calculated using typical daily load curves and annual operational statistics derived from long-term historical data, which are representative of the actual operating conditions of the distribution network.

At the node level, several indicators are proposed to evaluate the potential and effectiveness of energy storage deployment at individual buses, including the node optimization potential indicator (O_1), economic efficiency indicator (O_2), and wind power integration enhancement rate indicator (O_3). These indicators are calculated using node-level operating data and power flow results associated with the original network topology. At the block level, indicators such as the block matching rate improvement (L_1) and high-quality demand improvement rate (L_2) are introduced to quantify the enhancement of regional source–load coordination and demand quality after energy storage deployment. Block-level indicators are obtained by aggregating the characteristics of nodes belonging to the same planning block. At the grid level, system-wide indicators including the high-quality demand improvement rate (G_1) and grid matching degree discreteness improvement (G_2) are used to evaluate the overall coordination and balancing performance of the distribution network. These indicators reflect the global impact of distributed energy storage deployment rather than localized effects. The calculation expressions for these indicators are presented in Formulas (31) to (37).

$$O_1 = \frac{1}{Z} \sum_{z=1}^Z \delta_z \quad (31)$$

$$O_2 = \frac{F_2}{F_1} \quad (32)$$

$$O_3 = \frac{\sum_{t \in T} (P_t^{\text{DG}} - P_t^{\text{EL}}) \cdot \Delta t}{\sum_{t \in T} P_t^{\text{DG}} \cdot \Delta t} - F_3 \quad (33)$$

$$L_1 = \sum_{w=1}^W \frac{\varphi_w - \varphi'_w}{\varphi_w} \quad (34)$$

$$L_2 = \sum_{w=1}^W \frac{\sum_{j \in Q_w} \sum_{z=1}^Z \delta_z}{\sum_{j \in C_w} \sum_{z=1}^Z \delta_z} \quad (35)$$

$$G_1 = \frac{\sum_{z=1}^Z \delta_z}{\sum_{z \in U} \delta_z} \quad (36)$$

$$G_2 = 1 - \frac{\sum_{u=1}^U (\varphi'_u - \bar{\varphi}')^2}{\sum_{u=1}^U (\varphi_u - \bar{\varphi})^2} \quad (37)$$

where δ_z is the average value of node quality indicators; Z is the total number of nodes configured with energy storage; φ_w and φ'_w is the degree of matching before and after the configuration of energy storage in the w -th block; L_1 and L_2 , as the evaluation indicators of the block dimension, contain W subindicators for evaluating a single block; W is the number of blocks configured with energy storage number; Cw is the set of nodes in the u -th block; Qw is the set of energy storage nodes in the w -th block; U is the set of grid nodes; φ_u and φ'_u is the matching degree before and after the u -th node is configured with energy storage; $\bar{\varphi}$ and $\bar{\varphi}'$ represent the average matching degree of the grid distribution prior to and subsequent to the setup of energy storage.

When indicators O_1 and O_3 approach 1, it shows that energy storage architecture improves power supply stability and grid flexibility to renewable energy variations, leading to increased operational resilience. $O_2 > 1$ demonstrates cost-effective storage that reduces costs and increases revenue, thus strengthening economic resilience. L_1 and L_2 values around 1 suggest better regional source-load balance and power supply quality through storage planning. The source-load balance has a direct impact on grid operational stability and resilience. Similarly, G_1 and $G_2 \approx 1$ show that energy storage planning improves grid-level source-load balance and power quality, boosting overall system resilience.

Case Settings and Results

An analysis was performed on the three-dimensional evaluation indicators of each configuration example to further assess their benefits. Figure 16 illustrates the evaluation indicators for each situation.

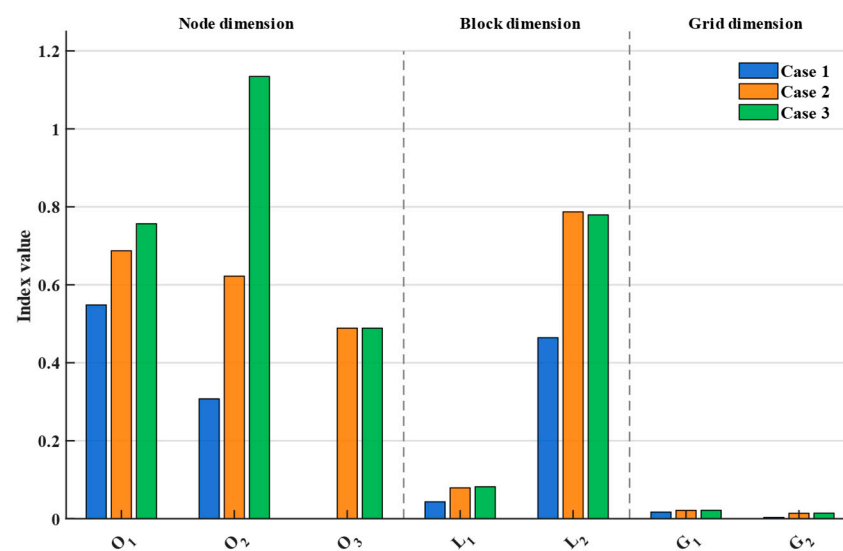


Figure 16. Comparison of evaluation indicators across cases.

By comparing the simulation results of Case 1 and Case 2, the optimization effect of the priority index on distribution network system stability and adaptive capability can be systematically quantified. Specifically, all key performance indicators demonstrate significant improvements in: In terms of O_1 , the indicator value of Case 2 increased from 0.548 in Case 1 to 0.684 (+25%), demonstrating that the priority index method can more effectively harness the potential to improve power supply quality at nodes while achieving superior load balancing, enhanced grid stability, and improved adaptability under uncer-

ainties. For O_2 , the metric value rose from 0.308 to 0.622 (+102%), demonstrating that the DESS configuration delivers superior economic benefits while maintaining stronger operational resilience, thereby enhancing the grid's responsiveness to external variations. In O_3 , Case 2 overcame Case 1's limitation in renewable utilization ($O_3 = 0$) through improved renewable-load coordination, enhancing grid adaptability to generation variability. In L_1 , the indicator value increased from 0.043 to 0.079 (+82%), demonstrating that Case 2 achieves more effective power distribution balancing within the block. The L_2 rose from 0.0464 to 0.787 (+70%), demonstrating the effectiveness of the prioritized index method in optimizing regional power supply quality and load-balancing. Additionally, Case 2 shows significant improvements in G_1 and G_2 . G_1 increased from 0.017 to 0.021 (+25%), and G_2 surged from 0.003 to 0.014 (+324%).

This indicates that the improvement in the internal grid's source-load balance has the greatest impact on enhancing the grid's resilience. In summary, the priority index approach markedly improves system resilience over exhaustive search methods by enhancing power supply quality and disturbance resistance at the node, block, and grid levels, while optimising source-load matching to ensure economic efficiency and operational flexibility.

Following the implementation of the sequential siting methodology in Case 3, all assessment metrics exhibit differing levels of enhancement relative to Case 2. This signifies an additional improvement in the system's disturbance management capabilities, while optimising the distribution of resilient resources, sustaining high absorption capacity, and enhancing the reliability of power supply quality. The disparity in source-load coordination among various grid blocks has diminished, leading to a more equitable distribution of overall resilience.

Among the three cases presented in Figure 12, Case 3 has the highest evaluation index value, signifying that it is the most rational option. DESS planning has enhanced grid economic efficiency through priority indices and sequential site selection methods, while boosting resilience against load fluctuations and renewable generation variability. This enables more stable and flexible grid operation.

6. Conclusions

The distributed energy storage-based distribution network resilience enhancement planning method proposed in this paper addresses the insufficient resilience of distribution networks caused by strong stochastic fluctuations in source-load dynamics, while also overcoming the significant limitations of traditional planning methods in terms of dynamic adaptability, multifunctional coordination, and global optimisation. This strategy improves both distribution network resilience and cost efficiency. The main contributions are as follows:

(1) A strategy for increasing distribution network resilience based on distributed energy storage is suggested, incorporating multidimensional functionality and demand assessment. The technique creates a priority index model at the grid-cell level, takes into account all relevant criteria, and provides systematic methodologies and tools for actual applications.

(2) A grid-based DESS planning method is proposed, featuring multi-objective sequential optimization to enhance distribution network dynamic adaptability and stability while achieving optimal load balancing and renewable energy integration.

(3) With only a 4% increase in investment costs, the proposed solution significantly improves operational resilience at all system levels, achieving a 27% improvement in power supply stability at critical nodes, increasing renewable energy integration from 0% to 48.8%, increasing block-level source-load matching by 88% and quality demand satisfaction by 68%, and improving grid-wide coordination uniformity by 324%. Simultaneously,

it provides significant economic benefits by offering a low-cost resilience enhancement solution for distribution networks with a high renewable energy penetration.

As the penetration of distributed generation and flexible loads continues to increase, enhancing resilience in distribution networks will benefit from coordinated planning across sources, loads, and storage. Building on the planning-oriented DESS deployment framework proposed in this paper, future research will investigate hierarchical source–load–storage coordination mechanisms, as well as their integration with more detailed operational and expansion models, to further improve system-wide self-balancing and resilience performance.

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